I/O LIST - ENGINE ROOM

Document IO-002 Rev A - repeated header, merged Type and Unit cells

				Range		Alarm limits				
Tag	**Description**	**Type**	**Unit**	**Low**	**High**	**HH**	**H**	**L**	**LL**	**Delay s**
ME-TAC1-TT03	Turning gear bearing temperature 3	AI	°C	0	150	144	128			5
AE-CLR1-TT01	Charge air cooler outlet temperature 1			0	120	105.6	96			5
AE-CLR1-TT02	Charge air cooler outlet temperature 2			0	120	115.2	96			30
AE-CLR1-TT03	Charge air cooler outlet temperature 3			0	120	110	90			10
AE-CLR1-TT04	Charge air cooler outlet temperature 4			0	120	115.2	102			30
AE-CLR1-PT01	Charge air pressure 1		bar	0	6			0.9		30
AE-CLR1-PT02	Charge air pressure 2			0	6			0.9		10
AE-CLR1-FT01	Cooler sea water flow		m3/h	0	400			100	44	15
AE-CLR1-XS01	Cooler pump 1 running	DI								
AE-CLR1-XS02	Cooler pump 2 running									
AE-AFS1-TT01	Auxiliary fuel temperature 1	AI	°C	0	160	147.2	136			0
AE-AFS1-TT02	Auxiliary fuel temperature 2			0	160	153.6	120			15
AE-AFS1-TT03	Auxiliary fuel temperature 3			0	160	154	136			10
AE-AFS1-PT01	Auxiliary fuel pressure 1		bar	0	16			2.4	1.3	5
AE-AFS1-PT02	Auxiliary fuel pressure 2			0	16			3.2	1.8	10
AE-AFS1-PT03	Auxiliary fuel pressure 3			0	16			2.4	1.3	2
AE-AFS1-FT01	Auxiliary fuel consumption 1		kg/h	0	800					
AE-AFS1-FT02	Auxiliary fuel consumption 2			0	800					
AE-AFS1-LT01	Auxiliary day tank level		%	0	100		75	15	11	5
AE-GBX1-TT01	Reduction gear bearing temperature 1		°C	0	130	114.4	97.5			15
AE-GBX1-TT02	Reduction gear bearing temperature 2			0	130	125	110			10
AE-GBX1-TT03	Reduction gear bearing temperature 3			0	130	124.8	104			10
AE-GBX1-TT04	Reduction gear bearing temperature 4			0	130	120	104			0
AE-GBX1-PT01	Reduction gear oil pressure 1		bar	0	10			2.5	0.8	10
AE-GBX1-PT02	Reduction gear oil pressure 2			0	10			1.5	0.5	2
AE-GBX1-VT01	Reduction gear vibration 1		mm/s	0	30	27.6	24			0
AE-GBX1-VT02	Reduction gear vibration 2			0	30	28.8	25.5			2
BLR-ECO1-TT01	Economiser gas temperature 1		°C	0	400	368	340			3
BLR-ECO1-TT02	Economiser gas temperature 2			0	400	368	300			5
BLR-ECO1-TT03	Economiser gas temperature 3			0	400	384	340			0

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BLR-ECO1-TT04	Economiser gas temperature 4	AI	°C	0	400	352	300			10
BLR-ECO1-PT01	Economiser water pressure 1		bar	0	20		15	3		10
BLR-ECO1-PT02	Economiser water pressure 2			0	20		15	3		5
BLR-ECO1-AT01	Flue gas oxygen content		%	0	25			3.8	2	10
BLR-ECO1-FT01	Economiser circulation flow		t/h	0	30				3.3	5
BLR-FDB1-LT01	Boiler feed tank level 1		%	0	100	88	80	20	11	5
BLR-FDB1-LT02	Boiler feed tank level 2			0	100	92	80	25	11	15
BLR-FDB1-LT03	Boiler feed tank level 3			0	100	88	80	20	11	0
BLR-FDB1-PT01	Feed pump discharge pressure 1		bar	0	30	27.6	24	6	3.3	30
BLR-FDB1-PT02	Feed pump discharge pressure 2			0	30	27.6	25.5	7.5	3.3	30
BLR-FDB1-TT01	Feed water temperature 1		°C	0	150		128	30		10
BLR-FDB1-TT02	Feed water temperature 2			0	150		120	30		3
BLR-FDB1-XA01	Feed pump 1 trip	DI								
BLR-FDB1-XA02	Feed pump 2 trip									
BLR-FDB1-XA03	Feed pump 3 trip									
BLR-BRN1-TT01	Burner flame temperature 1	AI	°C	0	1200	1056	900			0
BLR-BRN1-TT02	Burner flame temperature 2			0	1200	1056	960			0
BLR-BRN1-TT03	Burner flame temperature 3			0	1200	1152	1020			15
BLR-BRN1-PT01	Burner atomising pressure 1		mbar	0	100			25	11	15
BLR-BRN1-PT02	Burner atomising pressure 2			0	100			20	8	5
BLR-BRN1-ST01	Burner fan speed 1		rpm	0	3600	3456	2880			10
BLR-BRN1-ST02	Burner fan speed 2			0	3600	3168	3060			2
BLR-BRN1-FT01	Burner fuel flow 1		kg/h	0	1500.5		1200			0
BLR-BRN1-FT02	Burner fuel flow 2			0	1500.5		1125			0
BLR-BRN1-YS01	Burner 1 start command	DO								
BLR-BRN1-YS02	Burner 2 start command									
PMS-GRB1-VT01	Main busbar voltage section 1	AI	kV	0	12	11.5	10.2	2.4		15
PMS-GRB1-VT02	Main busbar voltage section 2			0	12	11.5	9.6	3		30
PMS-GRB1-VT03	Main busbar voltage section 3			0	12	10.6	9.6	2.4		5
PMS-GRB1-II01	Generator breaker current 1		A	0	3500	3220	2625			15

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PMS-GRB1-II02	Generator breaker current 2	AI	A	0	3500	3080	2975			30
PMS-GRB1-II03	Generator breaker current 3			0	3500	3360	2800			5
PMS-GRB1-II04	Generator breaker current 4			0	3500	3220	2800			10
PMS-GRB1-JT01	Generator active power 1		kW	0	5000	4400				10
PMS-GRB1-JT02	Generator active power 2			0	5000	4400				10
PMS-GRB1-JT03	Generator active power 3			0	5000	4600				2
PMS-GRB1-ST01	Busbar frequency 1		Hz	45	65	64.2	62	50	46	5
PMS-GRB1-ST02	Busbar frequency 2			45	65	64.2	62	50	46	10
PMS-GRB1-XS01	Generator breaker 1 closed	DI								
PMS-GRB1-XS02	Generator breaker 2 closed									
PMS-GRB1-XS03	Generator breaker 3 closed									
PMS-GRB1-XS04	Generator breaker 4 closed									
PMS-SHR1-JT01	Shore connection power 1	AI	kW	0	2500	2200				10
PMS-SHR1-JT02	Shore connection power 2			0	2500	2200				5
PMS-SHR1-II01	Shore connection current 1		A	0	1600	1536	1200			3
PMS-SHR1-II02	Shore connection current 2			0	1600	1472	1360			2
PMS-SHR1-II03	Shore connection current 3			0	1600	1472	1200			10
PMS-SHR1-VT01	Shore connection voltage 1		V	0	480	461	360	96	53	15
PMS-SHR1-VT02	Shore connection voltage 2			0	480	442	408	120	24	30
BWTS-UVR1-II01	UV reactor lamp current 1		A	0	30		22.5	7.5		3
BWTS-UVR1-II02	UV reactor lamp current 2			0	30		24	4.5		30
BWTS-UVR1-TT01	UV reactor water temperature 1		°C	0	80	70.4	60			3
BWTS-UVR1-TT02	UV reactor water temperature 2			0	80	76.8	60			10
BWTS-UVR1-PT01	UV reactor pressure 1		bar	0	10		8	1.5	0.8	10
BWTS-UVR1-PT02	UV reactor pressure 2			0	10		8.5	2.5	0.8	3
BWTS-UVR1-PT03	UV reactor pressure 3			0	10		8	2.5	1.1	10
BWTS-UVR1-FT01	Ballast treatment flow 1		m3/h	0	3000		2550			15
BWTS-UVR1-FT02	Ballast treatment flow 2			0	3000		2250			3
BWTS-UVR1-AT01	Residual oxidant		ppm	0	10		8			0
BWTS-FLT1-PT01	Filter differential pressure 1		bar	0	2	1.92	1.5			15

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BWTS-FLT1-PT02	Filter differential pressure 2	AI	bar	0	2	1.84	1.6			5
BWTS-FLT1-PT03	Filter differential pressure 3			0	2	1.84	1.7			2
BWTS-FLT1-XS01	Filter 1 backflush active	DI								
BWTS-FLT1-XS02	Filter 2 backflush active									
BWTS-FLT1-XS03	Filter 3 backflush active									
BWTS-FLT1-ZT01	Filter inlet valve 1 position	AI	%	0	100					
BWTS-FLT1-ZT02	Filter inlet valve 2 position			0	100					
TANK-TK21-LT01	Cargo oil tank 21 level 1			0	100	88	80	20	5	30
TANK-TK21-LT02	Cargo oil tank 21 level 2			0	100	92	75	15	8	5
TANK-TK21-LT03	Cargo oil tank 21 level 3			0	100	96	80	20	5	3
TANK-TK21-TT01	Cargo oil tank 21 temperature 1		°C	0	100		80			0
TANK-TK21-TT02	Cargo oil tank 21 temperature 2			0	100		85			30
TANK-TK21-PT01	Cargo oil tank 21 vapour pressure		mbar	-50	150		110	-10		10
TANK-TK22-LT01	Slop tank 22 level 1		%	0	100	96	80	25	8	30
TANK-TK22-LT02	Slop tank 22 level 2			0	100	88	85	15	8	5
TANK-TK22-LT03	Slop tank 22 level 3			0	100	92	80	20	5	10
TANK-TK22-TT01	Slop tank 22 temperature 1		°C	0	100		75			10
TANK-TK22-TT02	Slop tank 22 temperature 2			0	100		80			30
TANK-TK22-PT01	Slop tank 22 vapour pressure		mbar	-50	150		110	-20		0
TANK-TK23-LT01	Fresh water tank 23 level 1		%	0	100		75	20	5	5
TANK-TK23-LT02	Fresh water tank 23 level 2			0	100		80	15	5	30
TANK-TK23-LT03	Fresh water tank 23 level 3			0	100		85	15	5	5
TANK-TK23-LT04	Fresh water tank 23 level 4			0	100		80	15	5	30
TANK-TK23-TT01	Fresh water tank 23 temperature		°C	0	60		51			10
TANK-OWS1-AT01	Oil content monitor 1		ppm	0	30		22.5			3
TANK-OWS1-AT02	Oil content monitor 2			0	30		24			5
TANK-OWS1-LT01	Bilge holding tank level 1		%	0	100	92	80	25		10
TANK-OWS1-LT02	Bilge holding tank level 2			0	100	92	85	15		10
TANK-OWS1-FT01	Separator discharge flow		m3/h	0	5		4			5
TANK-OWS1-XA01	Separator alarm 1	DI								